PROPERTY RENTAL MANAGEMENT SYSTEM

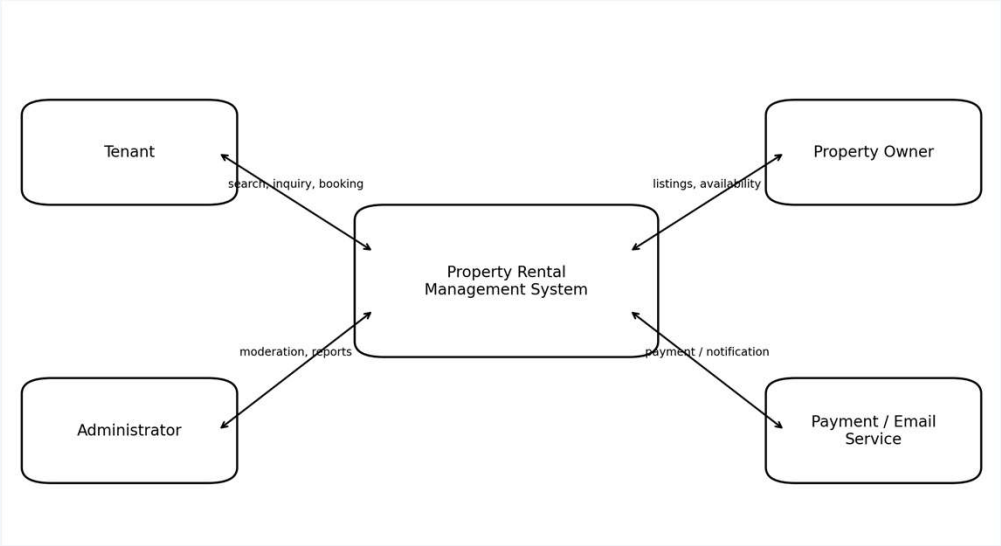
Software Quality Assurance Activity

SOFTWARE QUALITY ASSURANCE (23CSE701)

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Academic Year: 2026–2027

1. System Overview & Quality Objectives



Tenant

Search properties, view details and s

Owner

Manage listings, availability and rental reques

Administrator

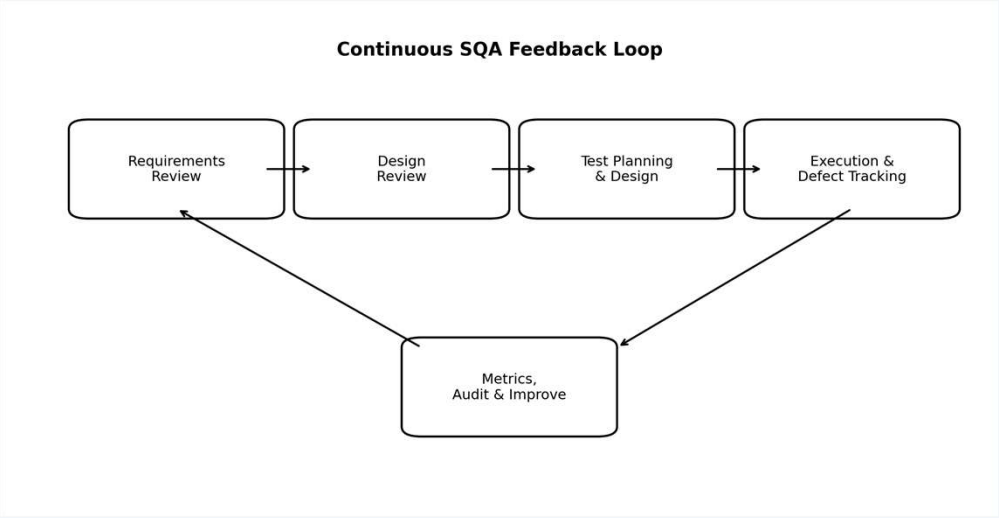
Moderate users/listings and review is

External Services

Payment and email are treated as dependencie

- Functional correctness
- Security and role-based access
- Reliable booking and availability
- Usable rental workflow

2. Overview of the SQA Plan



Requirements

Correct, complete, consistent and tes

Design

Architecture, data integrity and authorization.

Testing

Unit, integration, system, security an

Evidence

RTM, checklists, defects, metrics and documer

- Quality is planned across the lifecycle
- Prevention + detection + corrective action
- Evidence is traceable to requirements

3. Requirement Review & RTM

Req.	Requirement / Feature	Test Evidence
FR-01	User registration	TC-01, TC-02
FR-02	Authentication	TC-03, TC-04
FR-03	Property search	TC-05, TC-06
FR-05	Listing management	TC-07
FR-06	Availability rule	TC-08, TC-09
FR-07	Rental request	TC-10
FR-09	Administration	TC-11
FR-10	Status recording	TC-12

Review Checks

Unique IDs • testability • consistency • role permissions • error conditions

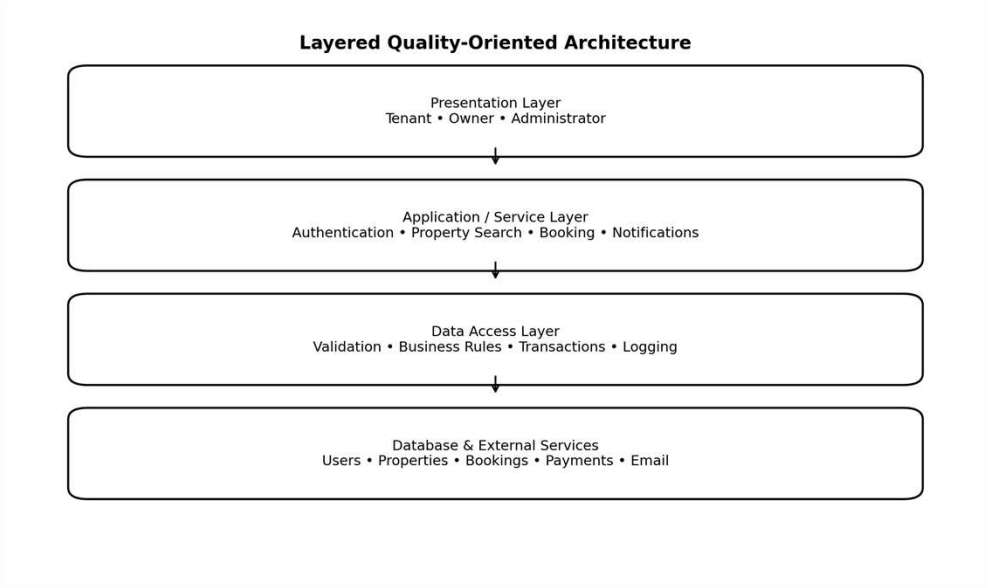
Key Finding

Performance expectations need a measurable target; critical requirements a

RTM Purpose

Connects requirements → design → test cases → evidence.

4. Design Review



Separation

Presentation, service, data and persis

Authorization

Protected operations enforce role checks.

Data Integrity

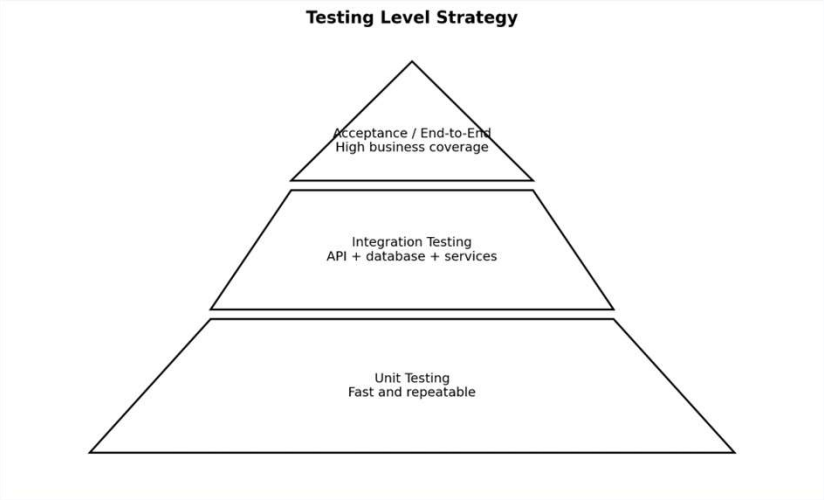
Booking and availability need transa

Failure Handling

External-service failures must not corrupt core

- Main risks: concurrency, authorization, inconsistent status updates and insufficient audit logging
- Conceptual ERD separates users, properties, bookings, payments and reviews

5. Testing Activities



Test ID	Scenario	Expected Result
TC-01	Valid registration	Account created
TC-04	Invalid login	Access denied safely
TC-05	Rent-range search	Matching properties only
TC-08	Book unavailable property	Booking blocked
TC-09	Concurrent booking	Only valid transition
TC-11	Tenant accesses admin	Request denied

Testing Focus

Boundary + negative tests + regression + security checks

6. Documentation Review & Defect Management

Documentation Review

Requirements, architecture, test cases, defects and changes

Defect Lifecycle

Record → reproduce → assign → fix → regression → ver

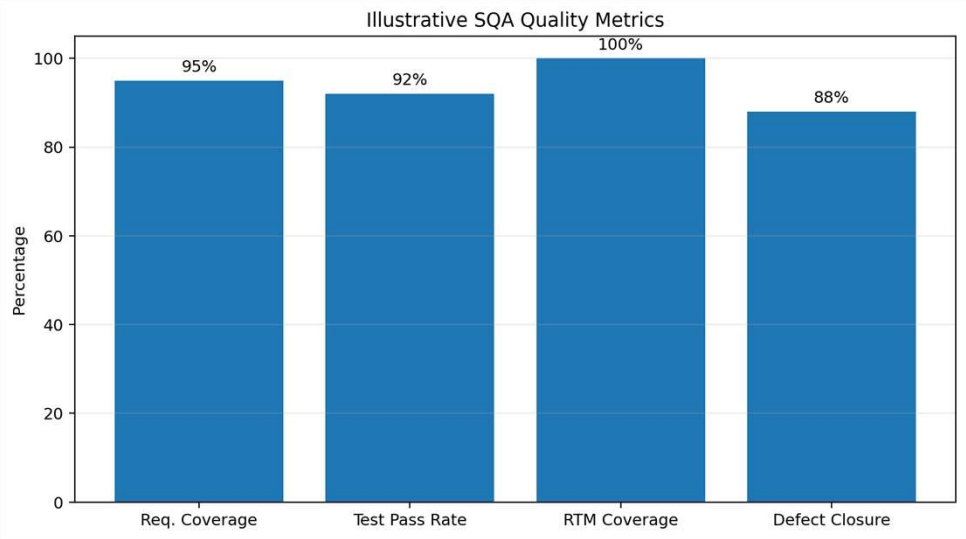
Document Control

Version, reviewer, date, change and approval status.

Defect	Area	Severity	Status
DEF-01	Availability	High	Verify
DEF-02	Authorization	Critical	Fixed
DEF-03	Validation	Medium	Fixed
DEF-05	Booking integrity	High	Action required

- Critical: missing role check can expose protected operations
- High risk: duplicate booking needs idempotency/concurrency control

7. SQA Metrics



Metric	Result	Meaning
Requirement Coverage	95%	Strong traceability
Test Pass Rate	92%	Most executed tests pass
RTM Coverage	100%	Requirements mapped
Defect Closure	87.5%	Some work remains
High/Critical Open	0 at gate	Release target
Review Action Closure	90%	Minor follow-up

Important

Values are illustrative academic SQA evidence, not production telemetry.

8. Observations, Risks & Findings

Risk	Impact	Primary Control
Unauthorized admin action	High	Role-based access tests
Duplicate/conflicting booking	High	Transactions + idempotency
External payment inconsistency	High	Failure/reconciliation tests
Incomplete audit evidence	Medium	Event logging
Poor search performance	Medium	Performance baseline

- Maintain RTM continuously
- Automate regression for critical workflows
- Define measurable performance baselines

Finding 1

Traceability improves verification and change-impact analysis.

Finding 2

State consistency is central to rental workflows.

Finding 3

Security, concurrency and external failures need targeted tests.

9. Conclusion & Key Takeaways

Requirement → Test

Critical requirements should have traceable design and te

Quality → Evidence

Checklists, defects and metrics make quality measurable

Risk → Action

Authorization, concurrency and data integrity need focused co

- Requirement Review: structured checks + RTM
- Design Review: architecture, authorization and transaction risks
- Testing: functional, integration, security, usability and regression
- Documentation Review: consistency and change control
- Metrics: repeatable evidence for release-quality decisions

References used in the report: ISO/IEC 25010 • ISO/IEC/IEEE 29119 • ISO/IEC 20246 • ISO/IEC/IEEE 12207 • IEEE 730 • IEEE 1012 • ISO 9001 • ISO/IEC 27001 • ISTQB CTFL • OWASP ASVS